

Tutorial Using the STM32F429 evaluation board

11 Sep 2024



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1. STM32F429 Overview



STM32F429 board

1. STM32F429 Overview

STM32F429 Features:

- Arm® Cortex®-M4 core-based microcontroller
- Reset and user push-buttons
- Six LEDs:
 - LD1 (red/green) for USB communication
 - LD2 (red) for 3.3 V power-on
 - Two user LEDs: LD3 (green) and LD4 (red)
 - Two USB OTG LEDs: LD5 (green) V_{BUS} and LD6 (red) OC (overcurrent)
- Communication interface: UART, SPI, CAN,...
- Timer, GPIO, PWM, ADC,...
- NVIC (Nested vectored interrupt controller)

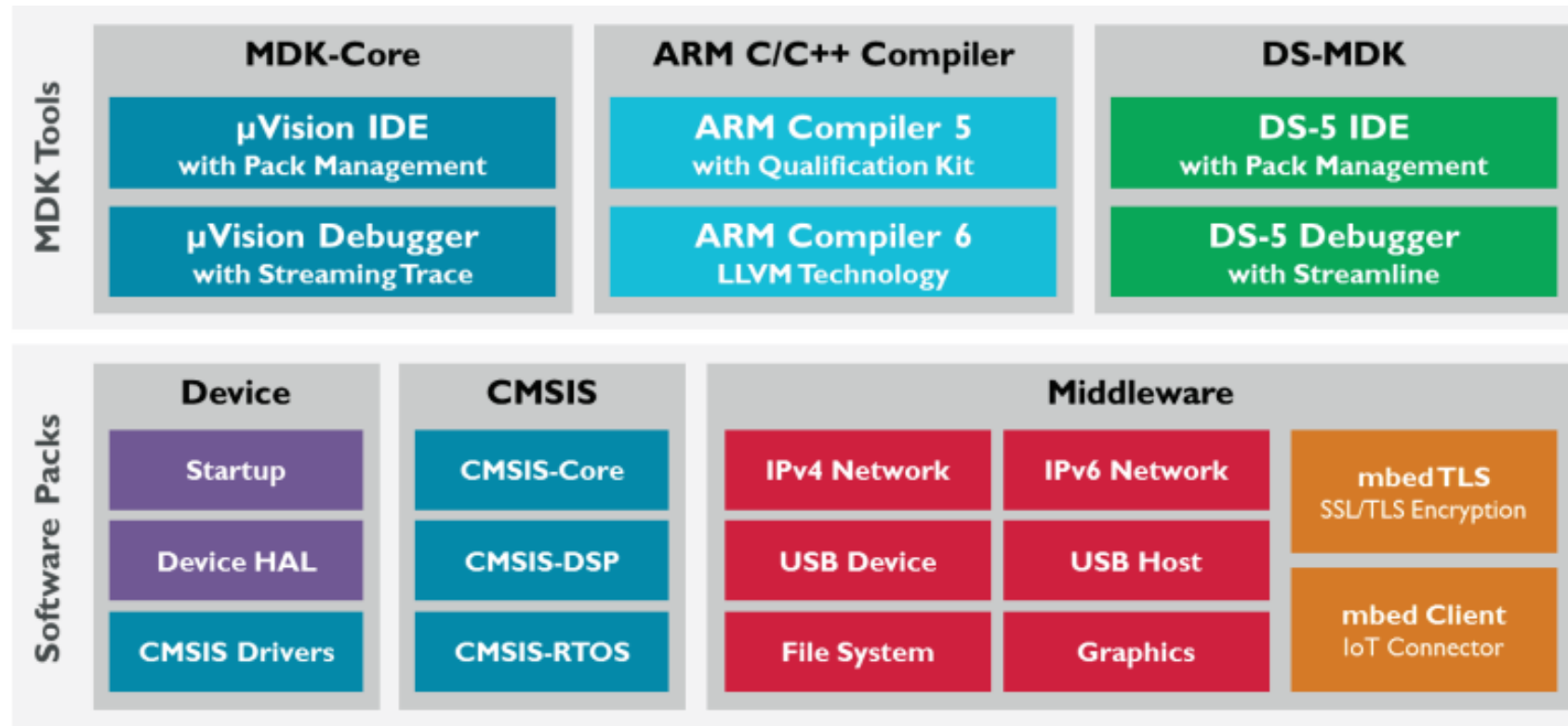
Board connectors:
USB Micro-AB



2. Development Tool chain Overview (KEIL)

MDK Microcontroller Development Kit

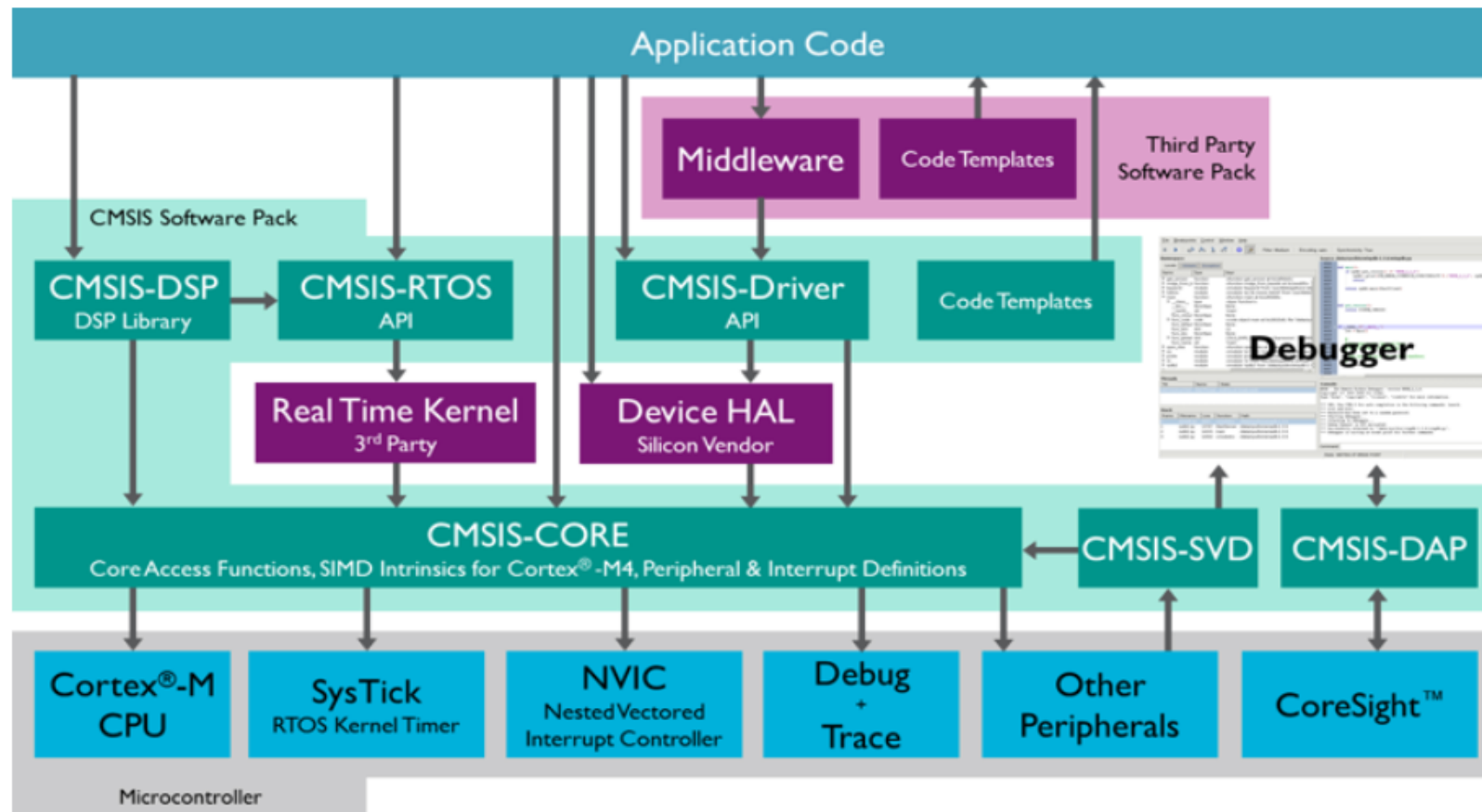
Keil® MDK is the most comprehensive software development solution for Arm®-based microcontrollers and includes all components that you need to create, build, and debug embedded applications.



2. Development Tool chain Overview (KEIL)

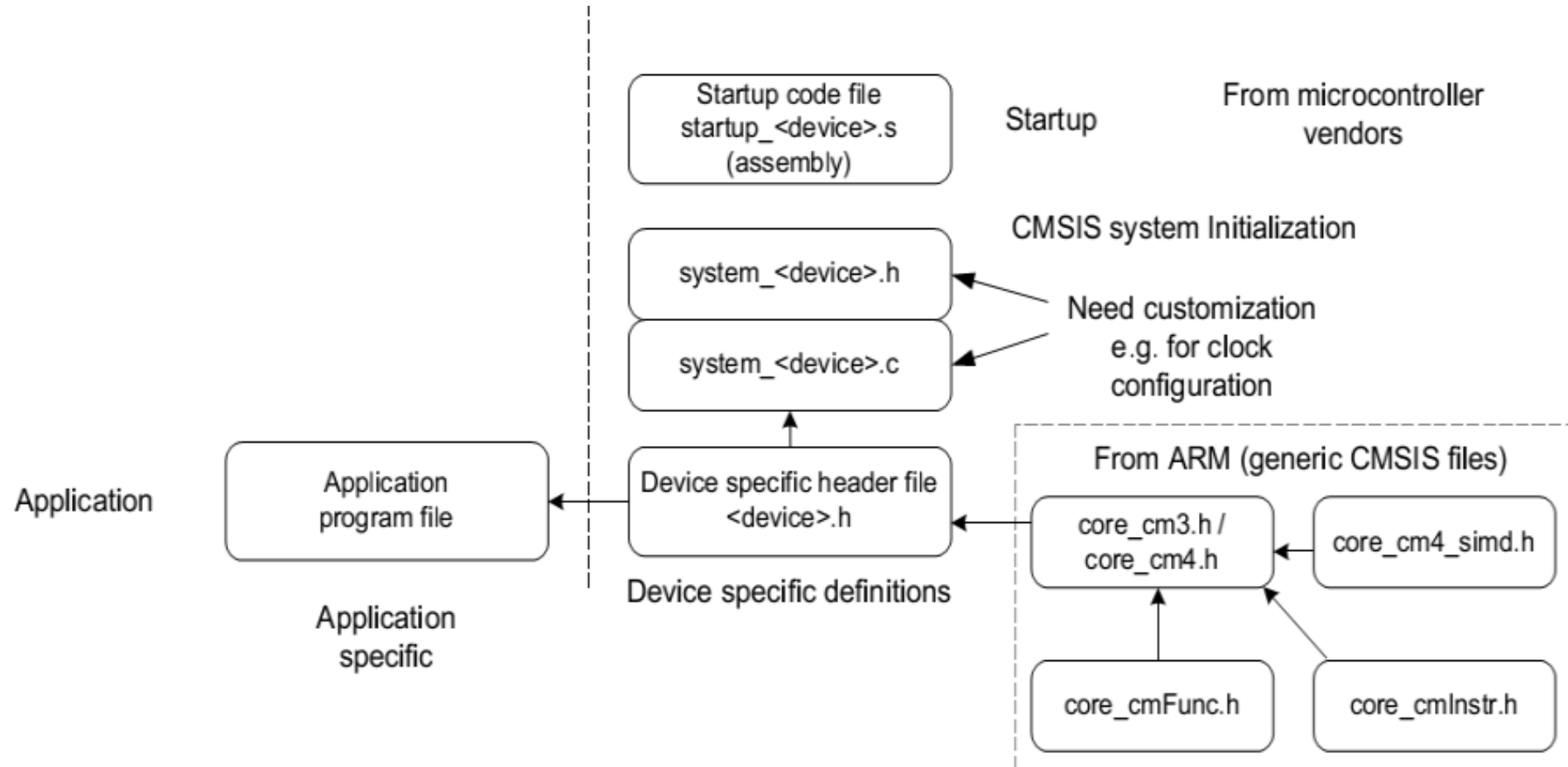
Common Microcontroller Software Interface Standard (CMSIS)

- ❖ **CMSIS** is a vendor-independent abstraction layer for microcontrollers that are based on Arm Cortex processors.
- ❖ **CMSIS** provides interfaces to processor and peripherals, real-time operating systems, and middleware components.



2. Development Tool chain Overview (KEIL)

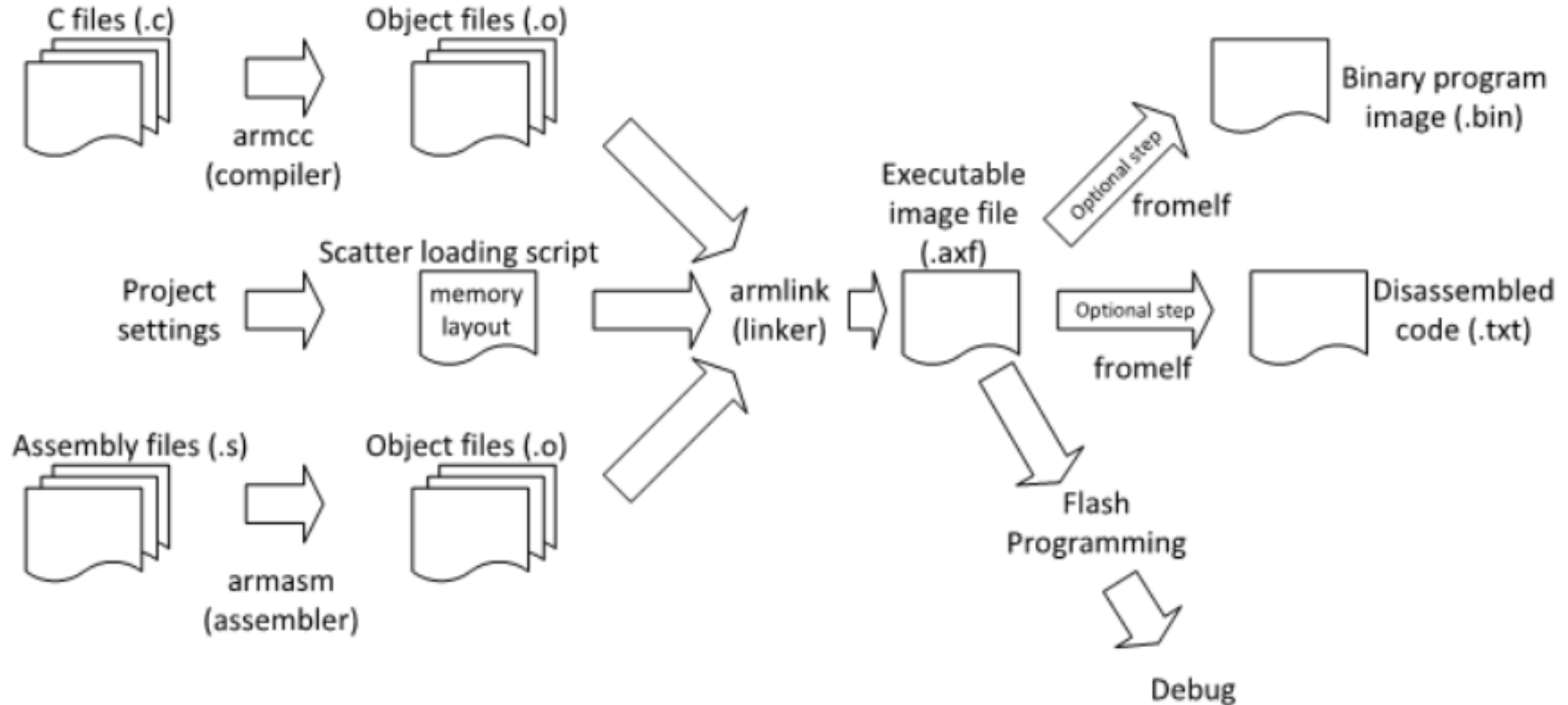
CMSIS Overview



Example project view when including CMSIS-Core files from ARM

2. Development Tool chain Overview (KEIL)

Typical program compilation flow



3. STM32F429 Getting Started Project



Software Development



HOST



USB to Micro-USB

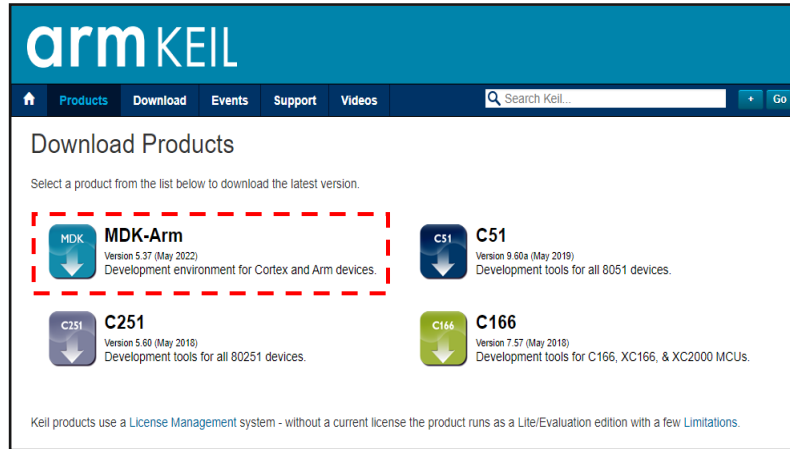


STM32F429 board

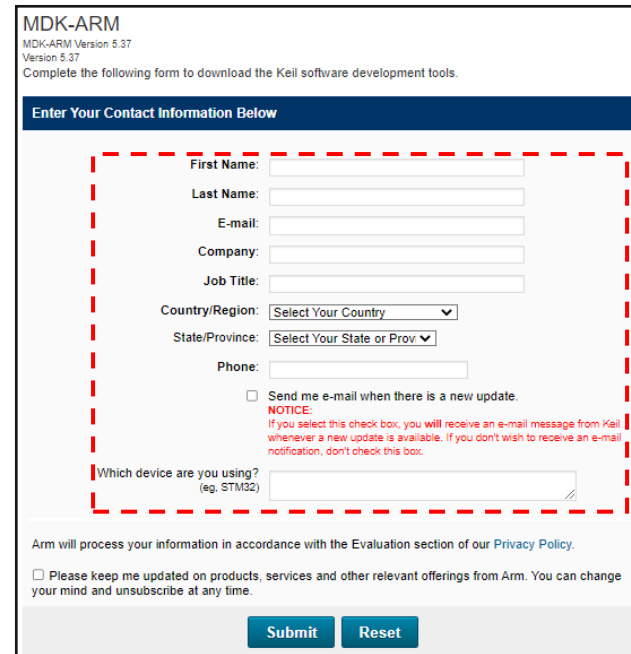
Hardware connection

3.1. Download Keil C μVision

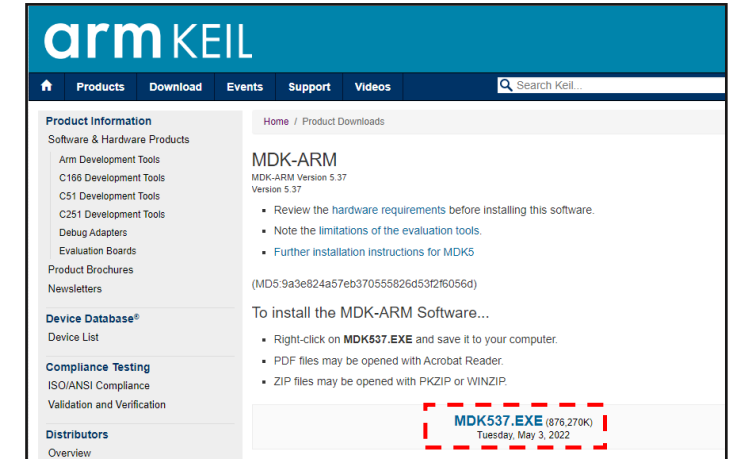
Link Download: <https://www.keil.com/download/product/>



Step 1: Click MDK-Arm

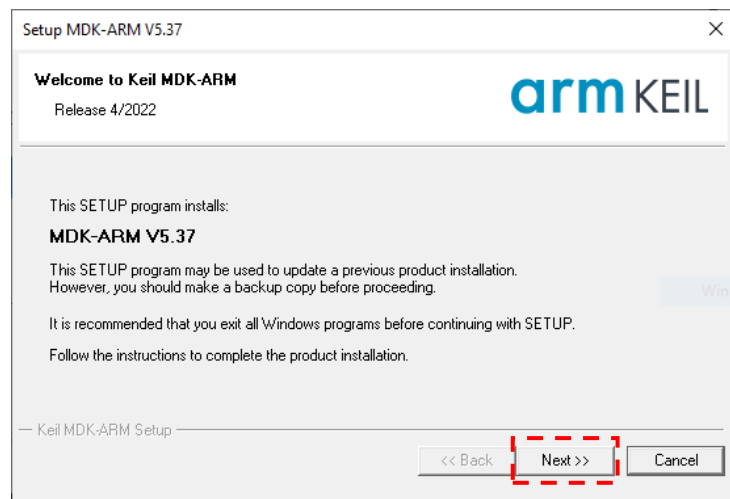


Step 2: Fill info and click Submit

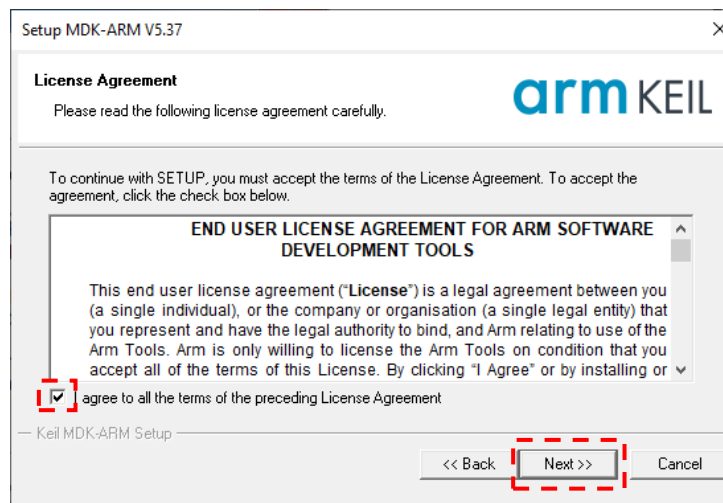


Step 3: Click MDKxxx.EXE

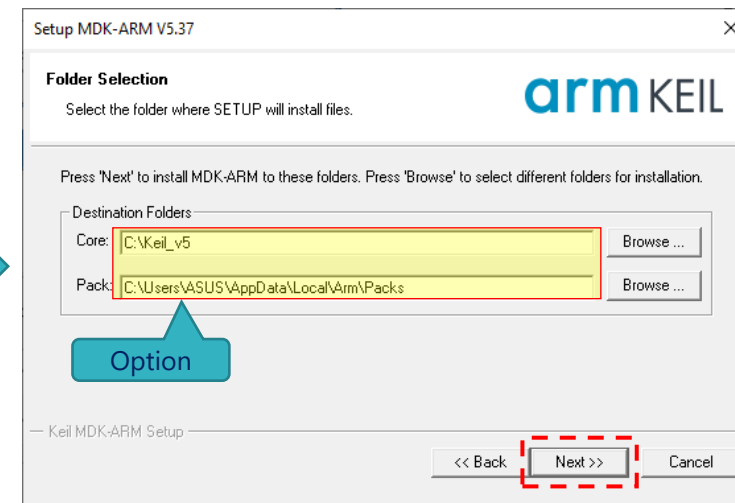
3.2. Install Keil C μ Vision



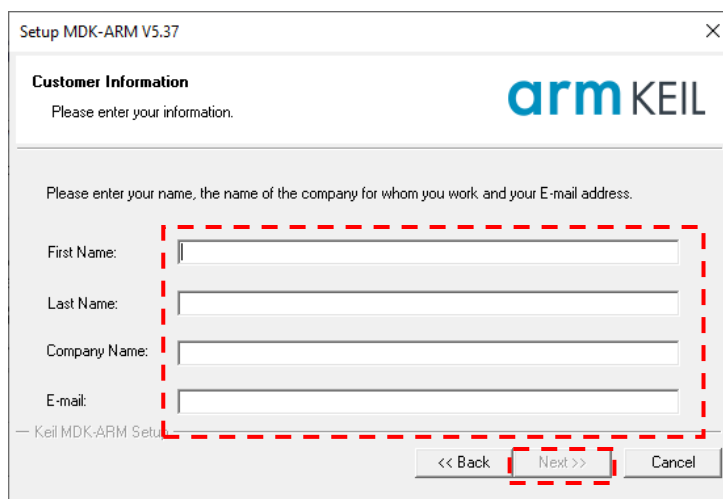
Step 1: Open MDKxxx.EXE and click Next



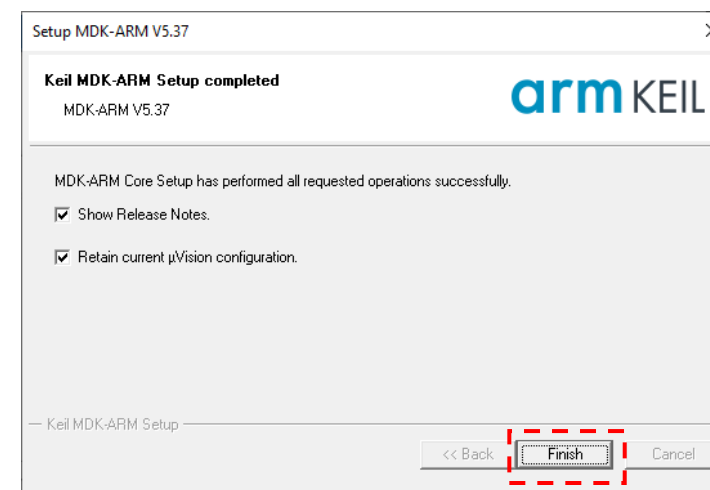
Step 2: Tick I agree and click Next



Step 3: Click Next



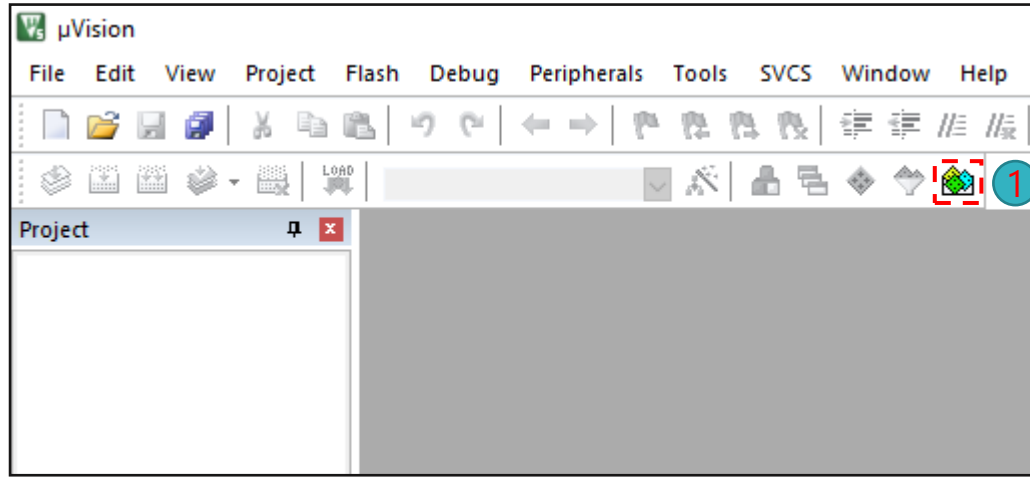
Step 4: Fill info and click Next to install



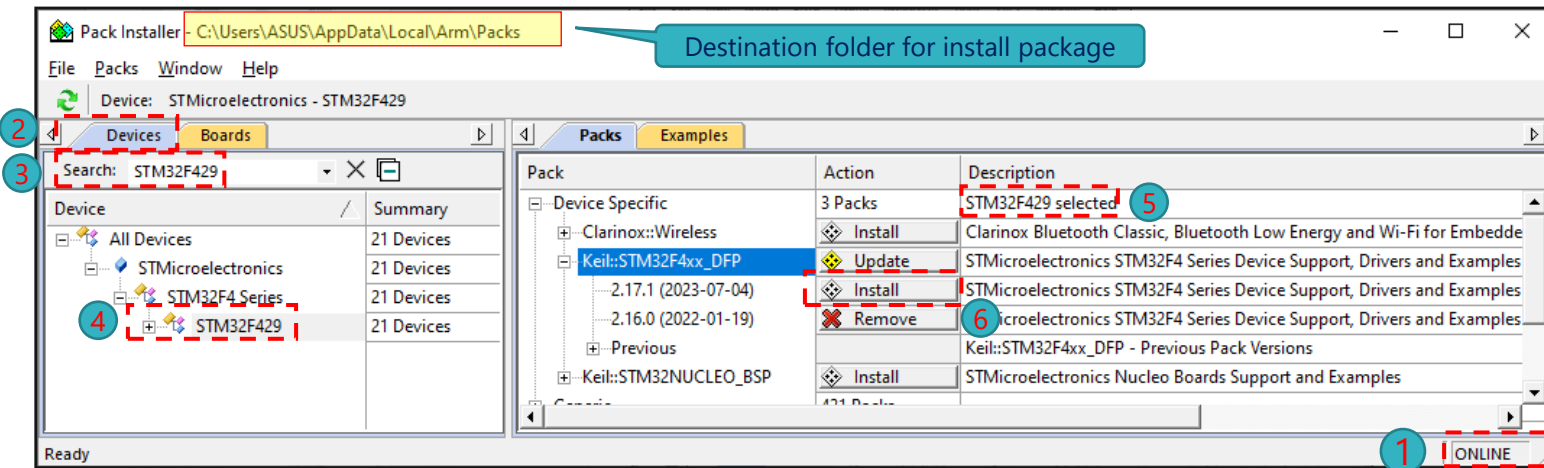
Step 5: Click Finish to end the installation

Wait

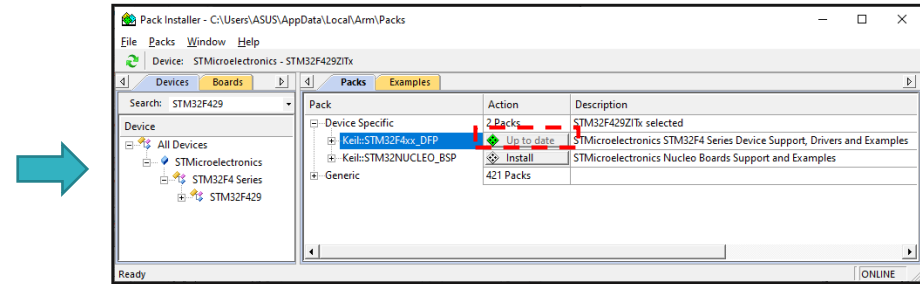
3.3. Install μVision Software Pack



Step 1: Start μVision and click Pack Installer:

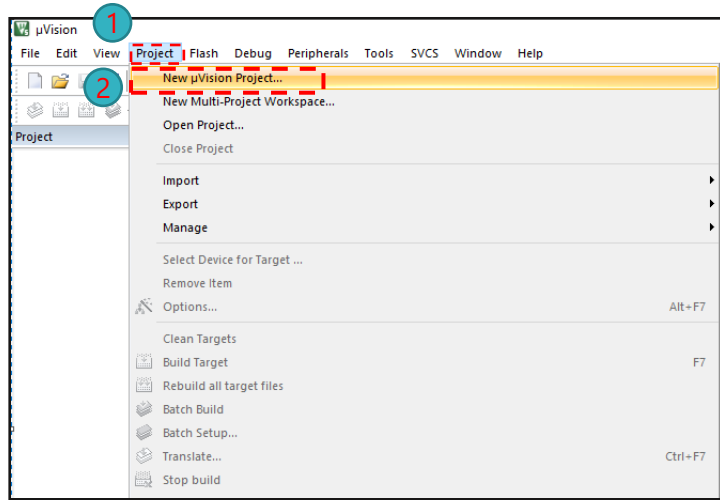


Step 2: Select STM32F429 as shown above and click Install

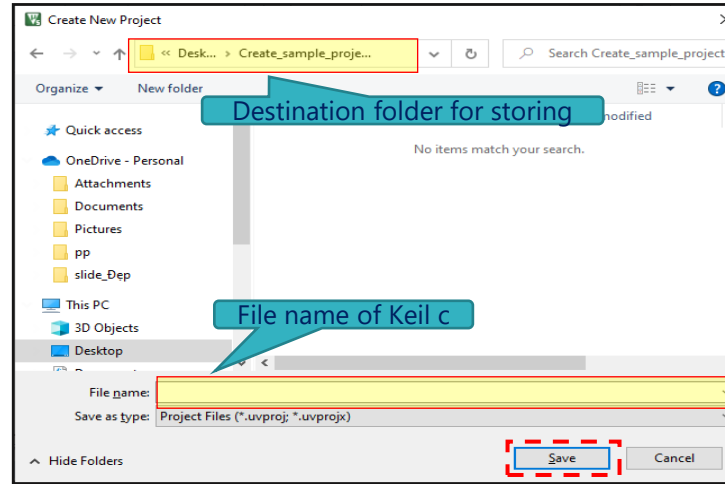


Install successful

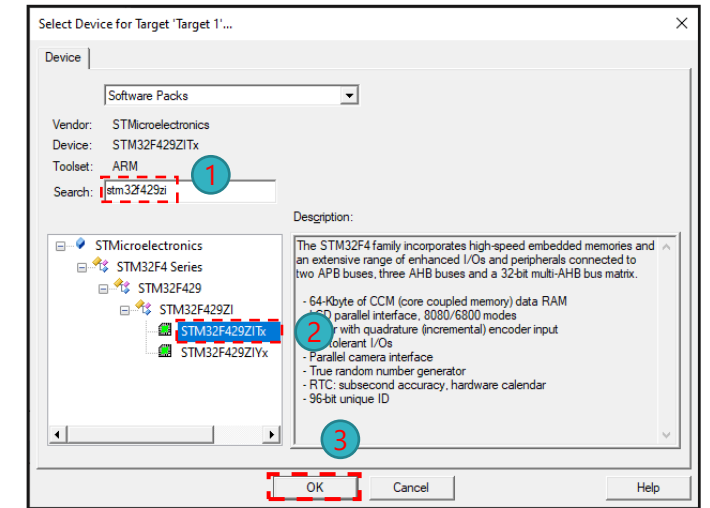
3.4. Create a new project



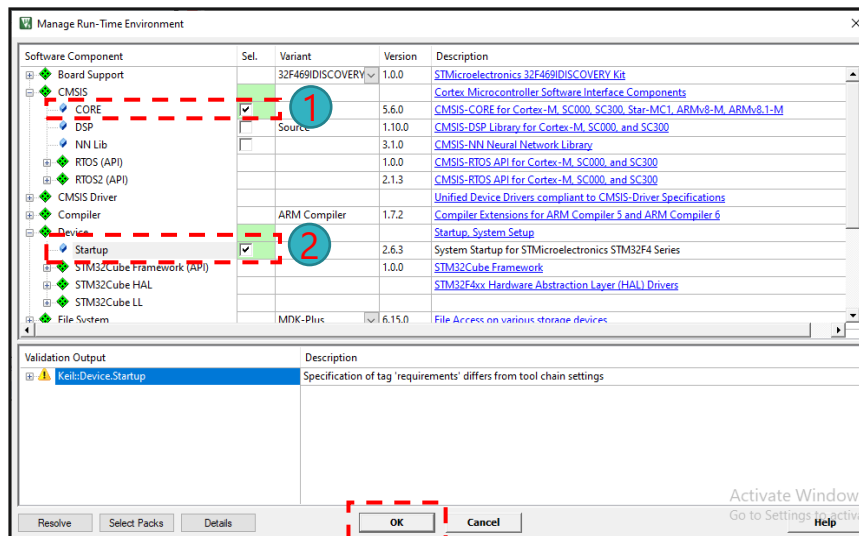
Step 1: Start uVision and click new project



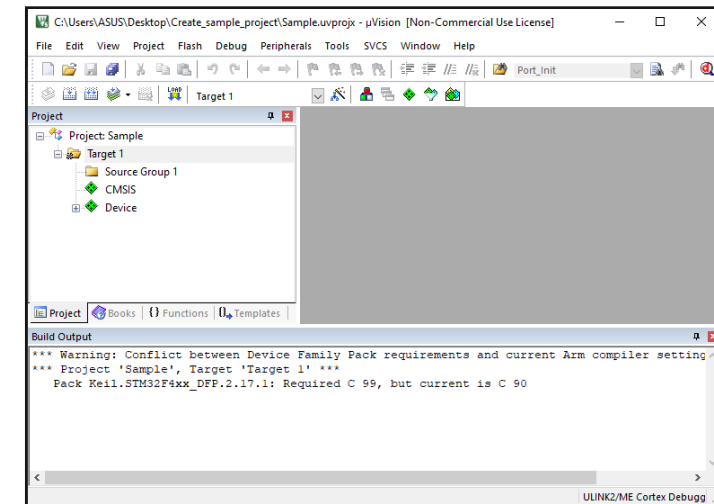
Step 2: Create file name and click Save



Step 3: Select device and click OK

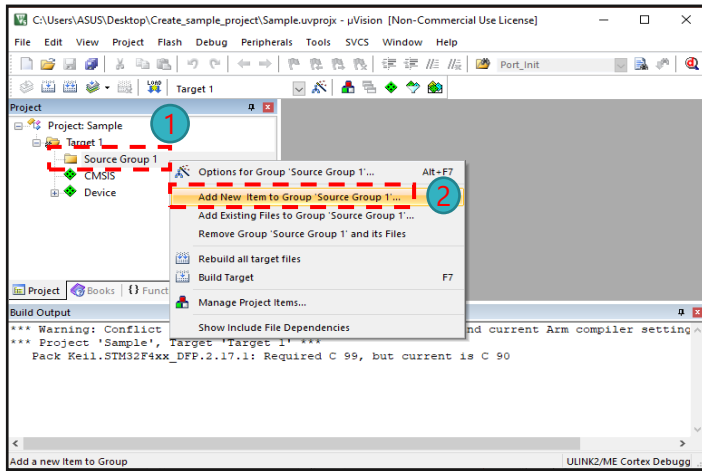


Step 4: Select as shown above and click OK

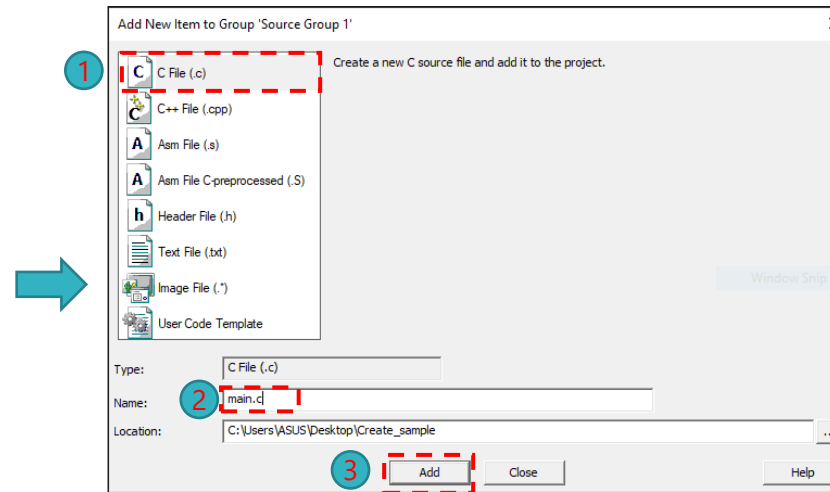


Create successful

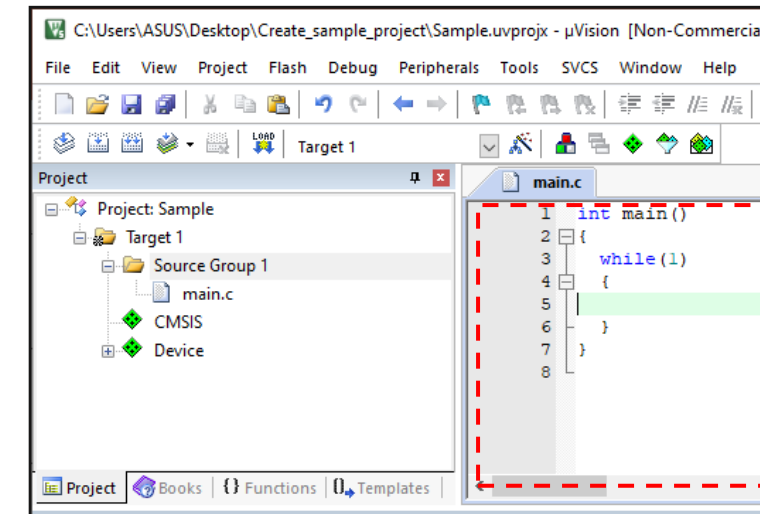
3.5. Setting parameter's project



Step 1: Right click Source Group 1

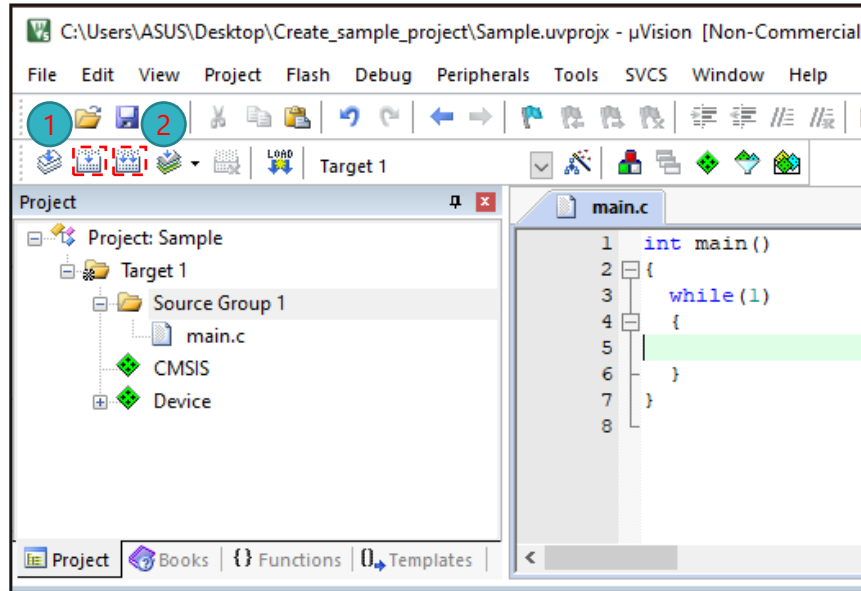


Step 2: Fill main.c and click Add

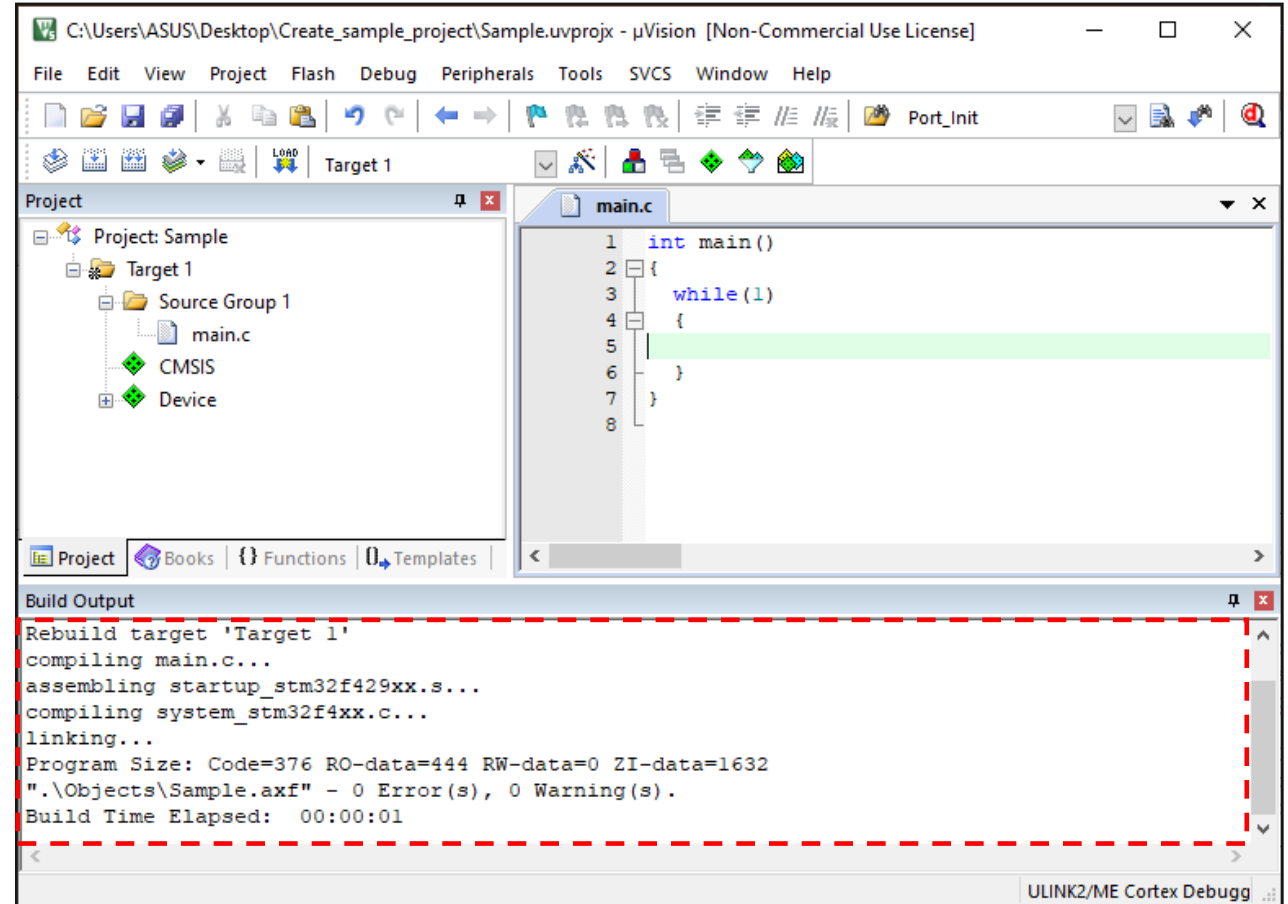


Step 3: Write main function

3.6. Build project

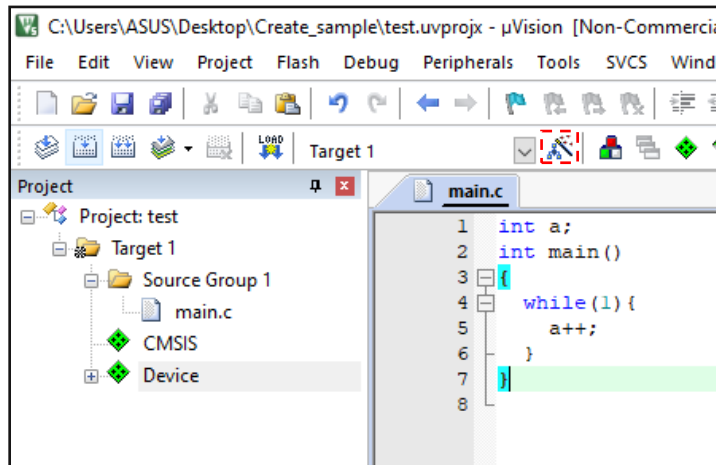


Step 1: Select Build or Rebuild

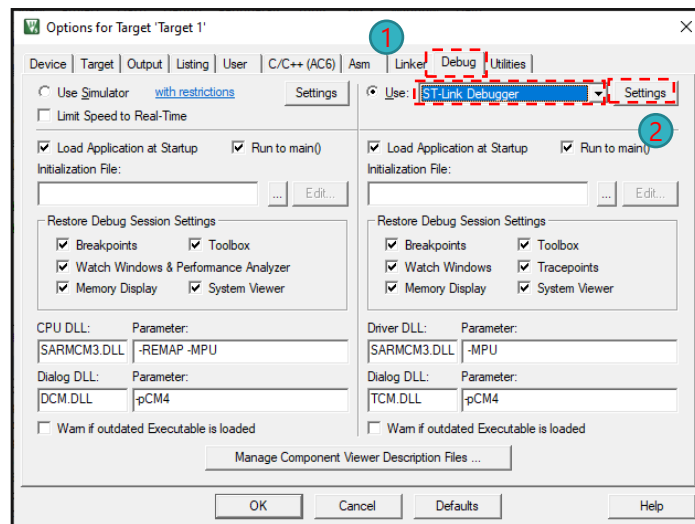


Build successful

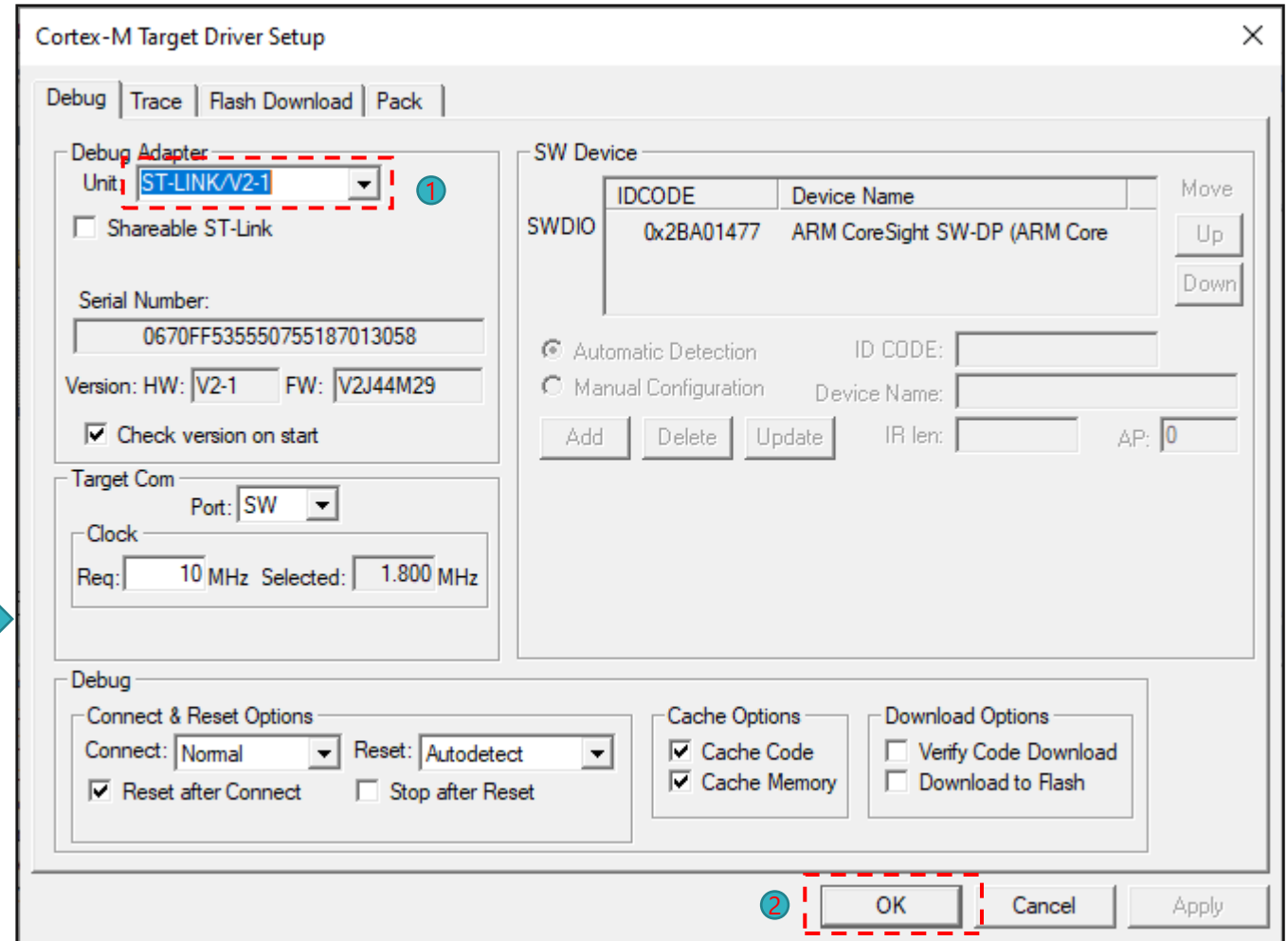
3.8. Configuring Debug



Step 1: Click Options for target

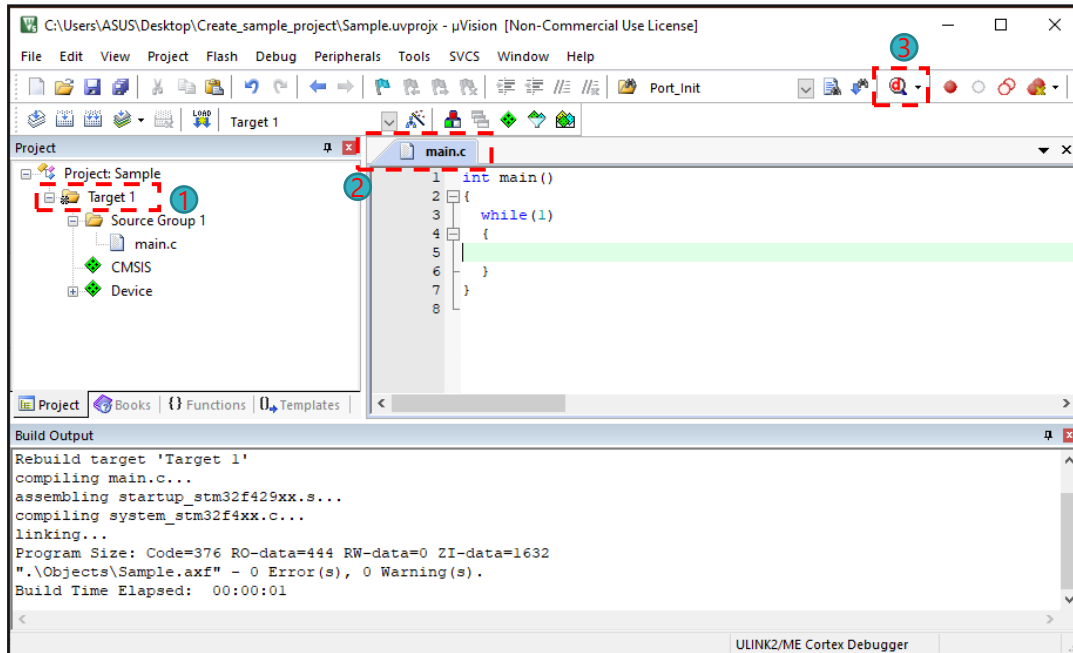


Step 2: Select as shown above and click Settings

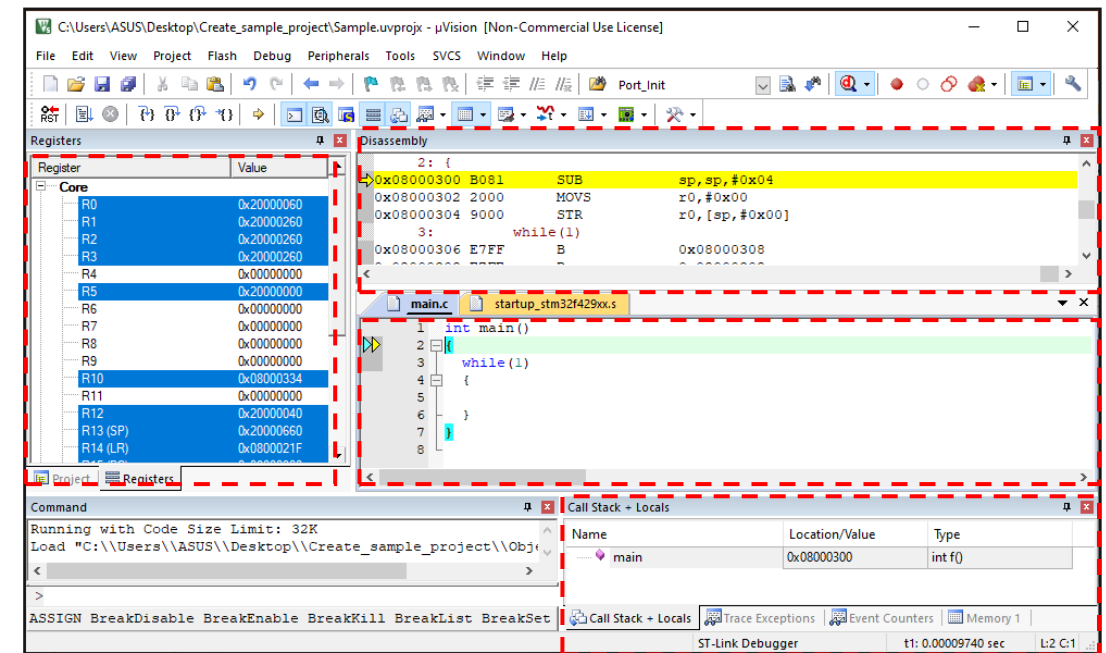


Step 3: Select as shown beside and click Ok
Note: The board must be connected

3.9. Debugging



Step 1: Click Start Debug Session(Ctrl+F5) as shown above



Step 2: Connect Debug successful

A nighttime cityscape featuring a large, brightly lit skyscraper in the center, which is highlighted by a large, semi-transparent circular inset. The inset shows the building's interior lights and its reflection in the water. The background shows other city buildings and a body of water under a dark sky.

Thank you